

Machine-checked binding (EasyCrypt, BindingViaCR.ec)
 $\text{MAL-BIND-K-}\{\text{CT, PK, CTX}\} \leq \text{CR}(\text{SHA3-256})$ — CDM Fig. 6, zero KEM assumption

specifies

ContextBound combiner — one Rust `no_std` core (the proven object)

5 faces

Rust

C ABI

WASM

Swift

Kotlin

compiled & run across

3 OS Linux · macOS · Windows-MSVC **5 ISA**
x86-64 · aarch64 · riscv64 · wasm32 · thumbv7em

all decapsulate to

byte-identical shared secret K (the *same* 32 bytes on every cell)

proof → *byte*

verified 6 ways: fixed KATs (X-Wing, RFC 7748) · NIST ACVP · independent differential · EasyCrypt proof · cross-platform byte-identity · property tests